

Anikait Kaushik

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International Rover Challenge Champion'2026 | International Space Drone Challenge First Runner-up'2026 | Research Intern @UQ | Winner – Best Mechanical Project ELC'24 | Mechatronics Engineering Student TIET'27 | Solidworks | Additive Manufacturing | MATLAB

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Education:

B.E / Btech in Mechatronics , Automation and Robotics Engineering

Expected Graduation May 2027

Thapar Institute of Engineering and Technology, Patiala, Punjab

- **CGPA** : 9.29/10.0 ([upto 5th Semester](#)) [**Previous Semester** : 10.0/10.0]
- **Position of Responsibility**: General Secretary of the Mechatronics and Robotics Society (MARS) and Team Captain of Team MARS at I.R.C

Skills

Programming Skills: C++ (Advanced), Python (Intermediate), C (Intermediate) ,

Software Packages: SolidWorks, Ansys, Ultimaker Cura, Arduino IDE, VS Code, MATLAB, AutoCAD, Creo Parametric, Altium Designer, Qualysis (QTM)

Technical Skills: Computer Aided Design (CAD), Simulation Analysis, Additive Manufacturing, PCB Designing (Basics), Finite Element Analysis (FEA) (Basics)

Non Technical Skills : Digital Media Marketing, Event Management, Financial Management, Documentation, Lead Generation, Logistics

Manufacturing and Fabrication Skills: Soldering, Welding, CNC Machining, 3D Printing

Language Proficiency: English (SRW), Hindi (SRW)

Research and Projects

Lower Limb Exoskeleton (Research Intern - University of Queensland)

June 2025 - Current

- Mentors: Dr. Ashish Singla
- Built a lower limb exoskeleton system to improve mobility and support rehabilitation, featuring real-time motion control,
- Fabricated and optimized it using SOLIDWORKS , simulating human and exoskeleton limbs for improved accuracy.
- Utilized GAIT laboratory data to thoroughly analyze gait cycles, providing detailed insights into COP and GRF metrics.
- Skills: Hardware, Electronics, Kinematics, CAD design, FEA, 3D printing, and Manufacturing.
- Tools: SolidWorks, ANSYS, Flashprint, Qualysis Track Manager (QTM)

International Space Drone Challenge 2026 ([Link](#))

March 2025 - Feb 2026

- Mentors: Dr. Ashish Singla, Dr. Sachin Kansal
- Developing a quadcopter drone from Carbon Fiber chassis and additive manufacturing. Selected BLDC motors and propellers for the drone along with its thrust calculations.
- Skills: Project Management, Team Leadership, CAD design, FEA, 3D printing, and Manufacturing.
- Tools: SolidWorks, ANSYS, FlashPrint.

International Rover Challenge 2026 (Mars Rover IV- Ganga) ([Link](#))

March 2025 - Feb 2026

- Mentors: Dr. Ashish Singla, Dr. Sachin Kansal
- Developing a Mars Rover prototype. Heading the mechanical and astrobiology expedition departments. Optimized the weight of the rover along with use of the new PU wheels from Wheeliez Australia. Headed the department in redesigning a more stable, rigid, and efficient astrological expedition on board laboratory for soil testing.
- Skills: Project Management, Team Leadership, CAD design, FEA, 3D printing, and Manufacturing.
- Tools: SolidWorks, ANSYS, Creality Slicer

Modular Reconfigurable Robot

Feb 2025 - Aug 2025

- Mentors: Dr. Ashish Singla, Dr. Sachin Kansal
- Designed and developed a modular reconfigurable robot using interchangeable modules to adapt quickly for diverse tasks and environments and performing various moments according to conditions.
- Skills: Project Management, Team Leadership, CAD design, Documentation, 3D printing, and Manufacturing.
- Tools: SolidWorks, Ultimaker cura , Arduino IDE.

International Rover Challenge 2025 (Mars Rover III- Alkananda)

March 2024 - March 2025

- Mentors: Dr. Ashish Singla, Dr. Sachin Kansal
- Designed and fabricated the entire soil subsystem for the Mars rover Prototype, from soil extraction Mechanism to onboard rover chemical testing and result evaluation through sensor integration
- Skills: Team Leadership, CAD design, FEA, 3D printing, CNC Machining .

Autonomous Cleaning Robot (ELC Internship 2024)

June 2024- July 2024

- Mentors: Dr. Ashish Singla, Dr. RK Duvedi
- Developed an Autonomous cleaning Robot capable of cleaning floors inside a room through efficient vacuum system and obstacle avoidance. Designed the robot's mechanical structure, electronic circuit and handled all the team operations and finances. Recognized as one of the top 10 best projects.
- Skills: CAD design, FEA, 3D printing, Sheet Metal, Digital Electronics, Documentation.
- Tools: SolidWorks, Arduino IDE, VS Code, Latex

Administrative Responsibilities

General Secretary of Mechatronics and Robotics Society and Team Captain at I.R.C ([Link](#))

Aug 2025- Current

- Heading and mentoring a team of 100+ students who are enthusiast and passionate about robotics.
- Handling a Rs. 8,00,000 budget strategically allocating resources for hands-on projects, international and national competitions, conferences,, and workshops.
- Heading all the team operations including administration, technology, management, finances, documentation, hospitality, design, logistics, presentations and marketing.

- Strengthening the society’s foundation, technical expertise, resources, corporate relations and teamwork.
- Mentoring and heading a team of 50+ students in participating in International Rover Challenge 2026, International Space Drone Challenge 2026 and DD Robocon 2026.
- Mentoring and heading a team of 50+ students in hands-on mini projects such as self- balancing robot, gimble stand, 4-DOF robotic manipulator, Reconfigurable Robot and Cleaning Robot Gen-2 (MARVIN).

Team Lead during ELC Internship2025

June 2025- July 2025

- Led a team of 10 students in designing and controlling an Autonomous Cleaning Robot(V2) and Modular Reconfigurable Robot
- Headed teams overall operations - Documentation, Finance.

Core Member of Mechatronics and Robotics Society

Sept 2024- Aug 2025

- Responsible for Mechanical, Marketing and R&D.
- Have worked on autonomous cleaning robot applying my research and ideation skills to obtain topology optimisation for existing model.
- Secured a Monetary sponsorship deal for our society.

MARS Magazine 2024-25: [Magazine link](#)

Achievements and Scholarships

- 1st Position Internationally | Internationa Rover Challenge 2026 Challenge , SPROS , Manipal ([Link](#))
- 2nd Position Internationally | Internationa Space Drone Challenge 2026 Challenge , SPROS , Manipal ([Link](#))
- 2nd Position Internationally | International Rover Challenge 2025 Challenge, SPROS, Goa ([Link](#))
- Recongnised under top 10 Internship Projects | ELC Summer Internship 2025 ([Link](#))
- Merit Scholarship | Rs.75,450 in 1st Year | Thapar Institute of Engineering and Technology
- Merit Scholarship | Rs.141,000 in 2nd Year | Thapar Institute of Engineering and Technology ([Link](#))
- Merit Scholarship | Rs.179,400 in 3rd Year | Thapar Institute of Engineering and Technology. ([Link](#))

Certifications and Workshops

- **MATLAB Onramp** ([Link](#))
 - **Participation in 7th International Conference of The Robotics Society (Advances in Robotics)** ([Link](#))
 - **Dynamics of Robotic System** ([Link](#))

Participated in Dynamics of Robotic Systems Workshop held at IIT Jodhpur during Advances in Robotics 2025 conference.

 - **Practicing Robotics: A Case Study on Reconfigurable Robotics** ([Link](#))
 - **International Rover Challenge 2025** ([Link](#))
 - **ELC Internship 2024** ([Link](#))

Secured Runners Up achievement and 2nd position globally.

Recognized among the top 10 internship projects.
- Aug. 2025

July 2025

July 2025

March 2025

Feb. 2025

Aug. 2024